

Day 5–6: Conditional Statements

- if
- if-else
- nested if
- switch

Programs:

- Grade calculator
- Leap year
- Menu-driven programs

Day 5–6 : Conditional Statements

First Question

What is Conditional Statement?

Suppose...

Morning lechav.

Outside weather chusthav.

If rain → Umbrella teesukuntav.

Else → Umbrella vaddu.

Idi manam daily life lo chestam.

Programming lo kuda same.

Computer ki ela telustundi?

Conditions dwara.

Example

If marks > 35

Pass

Else

Fail

Computer thinks like this.

Condition true?

YES → Execute this

NO → Execute another block

Condition ante

True or False evaluate ayye expression.

Example

10 > 5

True

5 > 20

False

Decision Making

Java lo decisions teesukodaniki

1. if
2. if else
3. else if
4. nested if
5. switch

use chestam.

1. if Statement

Syntax

```
if(condition)
```

```
{
```

```
    statements;
```

```
}
```

Example

```
int age = 20;
```

```
if(age >=18)
```

```
{
```

```
    System.out.println("Eligible");
```

```
}
```

Output

Eligible

Computer internally ila think chestundi.

age>=18 ?

20>=18 ?

True

↓

Execute block

Another Example

```
int salary=50000;
```

```
if(salary>30000)
```

```
{
```

```
    System.out.println("Rich");
```

```
}
```

Output

Rich

Suppose

salary=10000;

Condition?

10000>30000

False

Output?

Nothing.

Because

if executes only when condition is TRUE.

Flowchart

Start



Condition



True?



Yes



Execute

↓

End

If false

Skip block

Important Rule

Without braces

```
if(age>18)
```

```
System.out.println("Eligible");
```

Valid.

But

```
if(age>18)
```

```
System.out.println("Eligible");
```

```
System.out.println("Welcome");
```

Only first statement belongs to if.

Second statement always executes.

Example

```
age=15
```

Output

Welcome

Eligible print avvadu.

Better Practice

Always use

```
{
```

```
}
```

if Example

```
int num=10;
```

```
if(num>0)
```

```
{
```

```
System.out.println("Positive");
```

```
}
```

Output

Positive

2. if else Statement

Suppose

Rain unda?

Yes

Umbrella.

Else

No umbrella.

Syntax

```
if(condition)
```

```
{
```

```
}
```

```
else
```

```
{
```

```
}
```

Example

```
int marks=25;
```

```
if(marks>=35)
{
System.out.println("Pass");
}
else
{
System.out.println("Fail");
}
```

Output

Fail

Computer

marks>=35 ?

25>=35

False

↓

Else block

Example

```
int age=16;
```

```
if(age>=18)
```

```
{
```

```
Vote
```

```
}
```

```
else
{
Cannot Vote
}
Output
Cannot Vote
```

Difference

```
if
Condition True
```

↓

Execute

False

↓

Nothing

if else

True

↓

First block

False

↓

Second block

One of the blocks definitely executes.

else if Ladder

Suppose

Marks

90+

A Grade

80+

B Grade

70+

C Grade

Else

Fail

Syntax

if(condition1)

{

}

else if(condition2)

{

}

else if(condition3)

{

}

else

{

}

Example

int marks=82;

if(marks>=90)

{

System.out.println("A");

}

else if(marks>=80)

{

System.out.println("B");

}

else if(marks>=70)

{

System.out.println("C");

}

else

{

System.out.println("Fail");

}

Output

B

Computer Execution

82>=90?

False



82>=80?

True



Print B



Stop

Remaining conditions check cheyyadu.

Common Mistake

marks=95

if(marks>=35)

Pass

else if(marks>=90)

A Grade

Output

Pass

Wrong.

Because first condition itself true.

Java stops.

Always highest condition first.

Correct

90

80

70

35

Else

Nested if

Meaning

if inside another if.

Example

College Admission

Age check

Then marks check

Syntax

if(condition1)

{

 if(condition2)

 {

 }

}

Example

```
int age=20;
```

```
int marks=80;
```

```
if(age>=18)
```

```
{
```

```
    if(marks>=60)
```

```
    {
```

```
        System.out.println("Eligible");
```

```
    }
```

```
}
```

Output

Eligible

Flow

Age?

↓

True?

↓

Marks?

↓

True?

↓

Eligible

Another Example

ATM

Card valid?

↓

Yes

↓

PIN correct?

↓

Yes

↓

Withdraw

Nested if.

Example

```
boolean card=true;
```

```
int pin=1234;
```

```
if(card)
{
    if(pin==1234)
    {
        System.out.println("Cash");
    }
}
```

Difference

Normal if

One condition

Nested if

Condition inside condition.

switch Statement

Suppose

Restaurant Menu

1 Idli

2 Dosa

3 Poori

4 Upma

User enters

2

Output

Dosa

if else use cheyyachu.

But

switch easy.

Syntax

```
switch(expression)
```

```
{
```

```
    case value:
```

```
        Statements;
```

```
        break;
```

```
    case value:
```

```
        Statements;
```

```
        break;
```

```
    default:
```

```
        Statements;
```

```
}
```

Example

```
int day=2;
```

```
switch(day)
```

```
{
```


case 1:

```
System.out.println("Monday");
```

```
break;
```

case 2:

```
System.out.println("Tuesday");
```

```
break;
```

case 3:

```
System.out.println("Wednesday");
```

```
break;
```

default:

```
System.out.println("Invalid");
```

```
}
```

Output

Tuesday

Computer

day=2

↓

Check Case1

No

↓

Check Case2

Yes

↓

Execute

↓

break

↓

Stop

What is break?

Most Important Question.

Without break

Example

day=2;

switch(day)

{

case 1:

Monday

case 2:

Tuesday

case 3:

Wednesday

default:

Invalid

}

Output

Tuesday

Wednesday

Invalid

Why?

Because

Without break

Java enters matching case.

Then keeps executing remaining cases.

This is called

Fall Through

Remember this word.

With break

Case matched

↓

Execute

↓

break

↓

Exit switch

default

Suppose

case 1

case2

case3

Input

9

No match.

Then

default

executes.

Like

else.

Which Data Types Work in switch?

Allowed

byte

short

int

char

String

enum

Not Allowed

float

double

boolean

Example

```
String colour="Red";
```

```
switch(colour)
```

```
{
```

```
case "Red":
```

```
System.out.println("Stop");
```

```
break;
```

```
case "Green":
```

```
System.out.println("Go");
```

```
break;
```

}

Difference

if

Works with

>,<,>=,<=

switch

Works only

Exact match.

Example

switch(age)

case18

case20

Cannot

case age>18

Invalid.

if vs switch

if

Complex conditions

Supports >,<

Slower if many conditions

Flexible

switch

Exact value checking

Doesn't support >,<

Faster for many exact matches

Cleaner menu code

Which One Should We Use?

Suppose

Marks>35

Use

if

Suppose

Menu

1

2

3

4

Use

switch

Program 1 : Grade Calculator

Problem

Student marks input tiskoni grade print cheyyali.

Logic

90–100 → A

80–89 → B

70–79 → C

60–69 → D

35–59 → Pass

0–34 → Fail

Else → Invalid

Code

```
import java.util.Scanner;

public class GradeCalculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Marks: ");
        int marks = sc.nextInt();

        if (marks < 0 || marks > 100) {
            System.out.println("Invalid Marks");
        } else if (marks >= 90) {
            System.out.println("Grade A");
        } else if (marks >= 80) {
            System.out.println("Grade B");
        } else if (marks >= 70) {
            System.out.println("Grade C");
        } else if (marks >= 60) {
            System.out.println("Grade D");
        } else if (marks >= 35) {
            System.out.println("Pass");
        } else {
            System.out.println("Fail");
        }

        sc.close();
    }
}
```

Dry Run

Input:

82

Checks:

$82 < 0 \parallel 82 > 100 \rightarrow \text{False}$

$82 \geq 90 \rightarrow \text{False}$

$82 \geq 80 \rightarrow \text{True}$

Output:

Grade B

Program 2 : Leap Year

Rule

Year leap year avvali ante:

- 400 tho divisible ayithe $\rightarrow$ Leap Year
- Lekapotha 100 tho divisible ayithe $\rightarrow$ Not Leap Year
- Lekapotha 4 tho divisible ayithe $\rightarrow$ Leap Year
- Otherwise $\rightarrow$ Not Leap Year

Code

```
import java.util.Scanner;
```

```
public class LeapYear {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter Year: ");  
        int year = sc.nextInt();  
  
        if (year % 400 == 0) {  
            System.out.println("Leap Year");  
        } else if (year % 100 == 0) {
```

```
        System.out.println("Not a Leap Year");
    } else if (year % 4 == 0) {
        System.out.println("Leap Year");
    } else {
        System.out.println("Not a Leap Year");
    }

    sc.close();
}
```

Dry Run

Input:

2024

$2024 \% 400 = 24 \rightarrow \text{False}$

$2024 \% 100 = 24 \rightarrow \text{False}$

$2024 \% 4 = 0 \rightarrow \text{True}$

Output:

Leap Year

Input:

1900

$1900 \% 400 \neq 0$

$1900 \% 100 = 0$

Output:

Not a Leap Year

Input:

2000

$2000 \% 400 = 0$

Output:

Leap Year

Program 3 : Menu-Driven Program

Problem

Menu:

1. Addition
2. Subtraction
3. Multiplication
4. Division

User option select chestadu.

Code

```
import java.util.Scanner;

public class MenuDriven {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("1. Addition");
        System.out.println("2. Subtraction");
        System.out.println("3. Multiplication");
        System.out.println("4. Division");

        System.out.print("Enter Choice: ");
        int choice = sc.nextInt();

        System.out.print("Enter First Number: ");
        int a = sc.nextInt();

        System.out.print("Enter Second Number: ");
        int b = sc.nextInt();
```

```
switch (choice) {  
    case 1:  
        System.out.println("Result = " + (a + b));  
        break;  
  
    case 2:  
        System.out.println("Result = " + (a - b));  
        break;  
  
    case 3:  
        System.out.println("Result = " + (a * b));  
        break;  
  
    case 4:  
        if (b != 0) {  
            System.out.println("Result = " + (a / b));  
        } else {  
            System.out.println("Division by zero is not allowed");  
        }  
        break;  
  
    default:  
        System.out.println("Invalid Choice");  
}  
  
sc.close();  
}  
}
```

Dry Run

Input:

Choice = 3

a = 10

b = 5

Flow:

switch(3)

↓

case 3 matched

↓

10 * 5

↓

50

↓

break

↓

End

Output:

Result = 50

□ Interview Questions (Must Prepare)

1. What is a conditional statement?

Conditional Statement ante **oka condition ni check chesi, dani result (True or False) batti program next action decide cheyadam.**

2. What is the difference between if and if-else?

if	if-else
Only checks one condition	Checks condition and provides alternative
Executes only if condition is true	Executes one block whether condition is true or false
False ayithe nothing happens	False ayithe else block executes

3. When should we use else if?

Multiple conditions check cheyyali anukunte **else if ladder** use chestam

4. What is a nested if?

if statement lo inkoka if statement unte danini Nested if antaru.

5. What is the difference between nested if and else if?

Nested if	else if
if inside another if	Multiple conditions in sequence
Second condition depends on first	Conditions are independent
Used for dependent decisions	Used for choosing one option among many

6. What is a switch statement?

switch statement exact value ni compare cheyadaniki use chestam.

7. Why do we use break in a switch?

break statement switch nundi immediate ga bayataki ravadaniki use chestam

8. What is **fall-through** in a switch?

break use cheyyakapothe matching case nundi kindaki unna cases anni execute avutayi.

9. What is the purpose of the default case?

default ante switch lo **else** laga work chestundi.

10. Which data types are supported in switch?

byte, short, int, char, String, enum

enum Level { LOW, MEDIUM, HIGH } enum: collection of constants

11. Why are float, double, and boolean not allowed in a traditional switch?

Because floating point numbers are **not always stored exactly** in binary memory.

Suppose $0.1 + 0.2 = 0.3$ sometimes stored as 0.30000004

12. When should you choose if instead of switch?

Greater than, less than, range, complex

13. Can switch compare ranges like marks > 90? No, only compares exact values

14. What happens if no condition is true in an if without an else?

Nothing happens. Program next statement ni execute chestundi

15. What happens if multiple else if conditions could be true?

Java **top nundi bottom** check chestundi.

First true condition execute ayyaka remaining conditions check cheyyadu.

Key Points to Remember

- if executes **only when the condition is true**.
- if-else guarantees **exactly one block** will execute.
- In an else if ladder, **the first true condition wins**, and the rest are skipped.
- Arrange conditions from **most specific/highest** to **lowest**.
- Nested if is used when **one decision depends on another**.
- switch is best for **exact value matching** (menus, options, days, months).
- Always use break in traditional switch unless you intentionally want **fall-through**.
- default acts like the final fallback (else).
- Use braces {} even for single statements—it improves readability and prevents bugs.

Practice Assignment

Write programs for:

1. Check whether a number is positive, negative, or zero.
2. Find the largest of three numbers.
3. Check whether a character is a vowel or consonant.
4. Print the day name based on a number (1–7) using switch.
5. Build a simple ATM menu (Balance, Deposit, Withdraw, Exit).
6. Check whether a student is eligible for placement ($\text{CGPA} \geq 7.0$ and no backlogs).
7. Calculate an electricity bill based on unit ranges using an else if ladder.